

NAME OF STUDENT: Muhammad Danish ID No: 470-2024

INPUT/OUTPUT INSTRUCTIONS AND INT 21H

Objectives:

- To understand data transfer and exchange using MOV and XCHG instructions.
- To work with DB, DW, DD variables and appropriate registers.
- To understand the difference between value and address using LEA.

Introduction to Input and Output

A program generally performs:

INPUT → PROCESSING → OUTPUT

In Assembly Language, basic input and output can be performed using DOS services through: INT 21H

INT 21H is a DOS interrupt used to perform different input/output and system operations. The required function number is placed in the AH register before executing INT 21H.

MOV AH, function_number
INT 21H

For example:

MOV AH, 01H
INT 21H

Important INT 21H Functions**Function Purpose**

01H	Take input with echo
02H	Display character
07H	Take input without echo
09H	Display string
4CH	Return control to DOS / Terminate program
0BH	Check standard input status
2AH	Get system date
2CH	Get system time

1. Function 01H — Character Input

Function 01H is used to take one character from the keyboard.

MOV AH, 01H

INT 21H

The entered character is returned in:

AL

The entered character is also displayed on the screen automatically.

Example

If the user enters: A

Then: AL = ASCII value of A

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2. Function 02H — Character Output

Function 02H is used to display one character.

The character must be placed in DL.

MOV DL, 'A'

MOV AH, 02H

INT 21H

Output: A

3. INT 21H — Function 07H

Function 07H is used to take one character from the keyboard without displaying it on the screen.

MOV AH, 07H

INT 21H

The entered character is returned in: **AL**

Example

MOV AH, 07H

INT 21H

If the user enters: A

The character is stored in:

AL = ASCII value of A

01H → Input character with echo

07H → Input character without echo

The original manual describes 07H as working like 01H except that the entered character is not displayed.

4. Function 09H — String Output

Function 09H is used to display a complete string.

The string must end with \$.

Example

.DATA

MSG DB 'Hello World'\$ (\$ marks the end of the string.)

To display the string:

MOV DX, OFFSET MSG (DX contains the address of the string.)

MOV AH, 09H (AH = 09H selects the string-output function.)

INT 21H

5. Function 4CH — Program Termination

Function 4CH is used to terminate the program and return control to DOS.

MOV AH, 4CH

INT 21H

6. INT 21H — Function 0BH

Function 0BH is used to check whether a character is available from the keyboard.

Syntax

MOV AH, 0BH

INT 21H

The result is returned in AL.

Return Values

AL = 00H (No character is available.)

AL = FFH (A character is available.)

Example

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MOV AH, 0BH
INT 21H

7. INT 21H — Function 2AH

Function 2AH is used to obtain the current system date.

Syntax

MOV AH, 2AH

INT 21H

The date information is returned in different registers:

Register Information

CX Year

DH Month

DL Day

AL Day of the week

Example

MOV AH, 2AH

INT 21H

8. INT 21H — Function 2CH

Function 2CH is used to obtain the current system time.

Syntax

MOV AH, 2CH

INT 21H

The time information is returned in:

Register Information

CH Hour

CL Minutes

DH Seconds

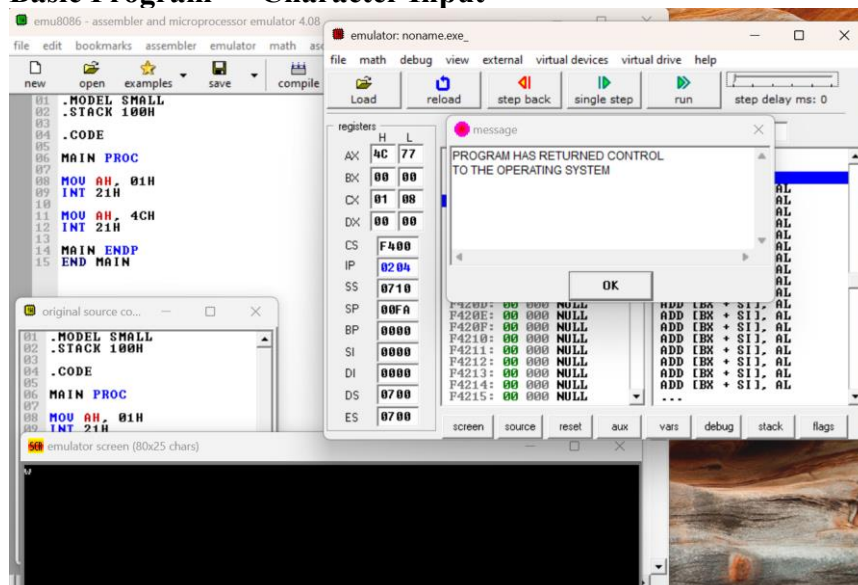
DL 1/100 second

Example

MOV AH, 2CH

INT 21H

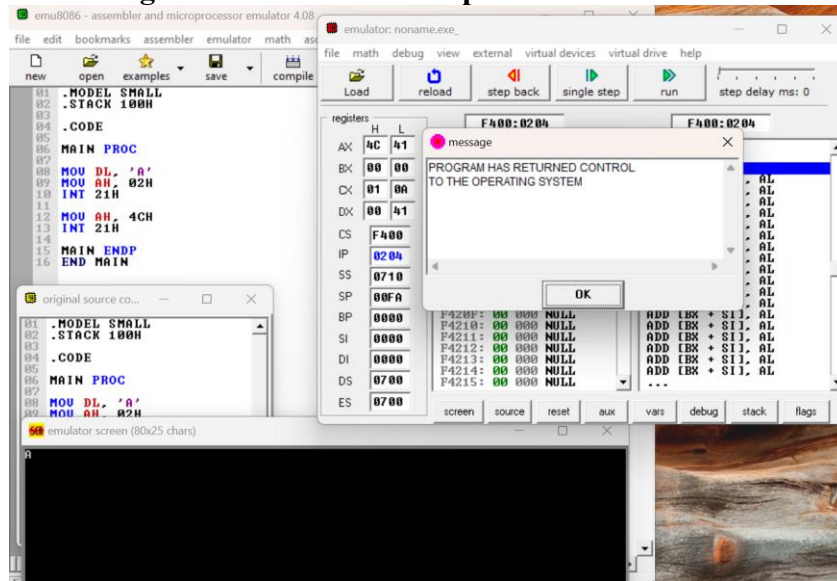
Basic Program — Character Input



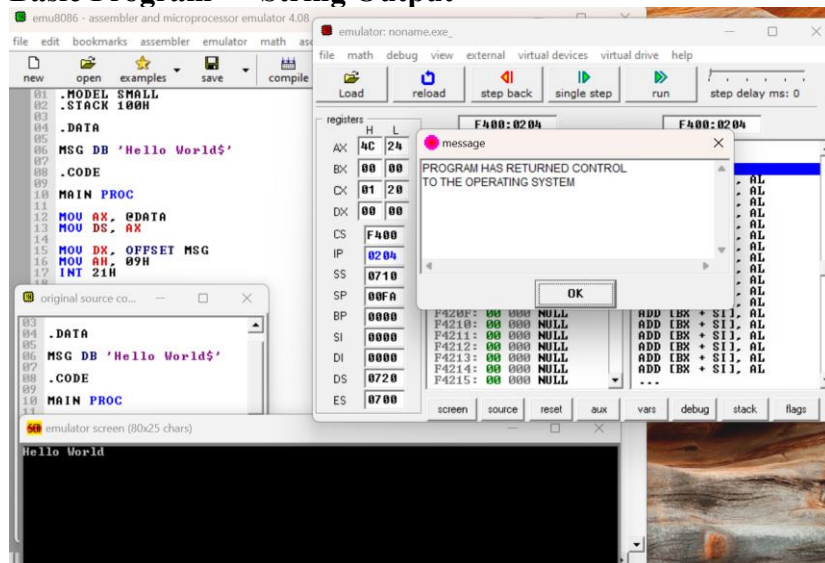
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Basic Program — Character Output



Basic Program — String Output



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Lab no 3 Task Solution

Task 1

Write a program to take one character from the user and display it on the screen.

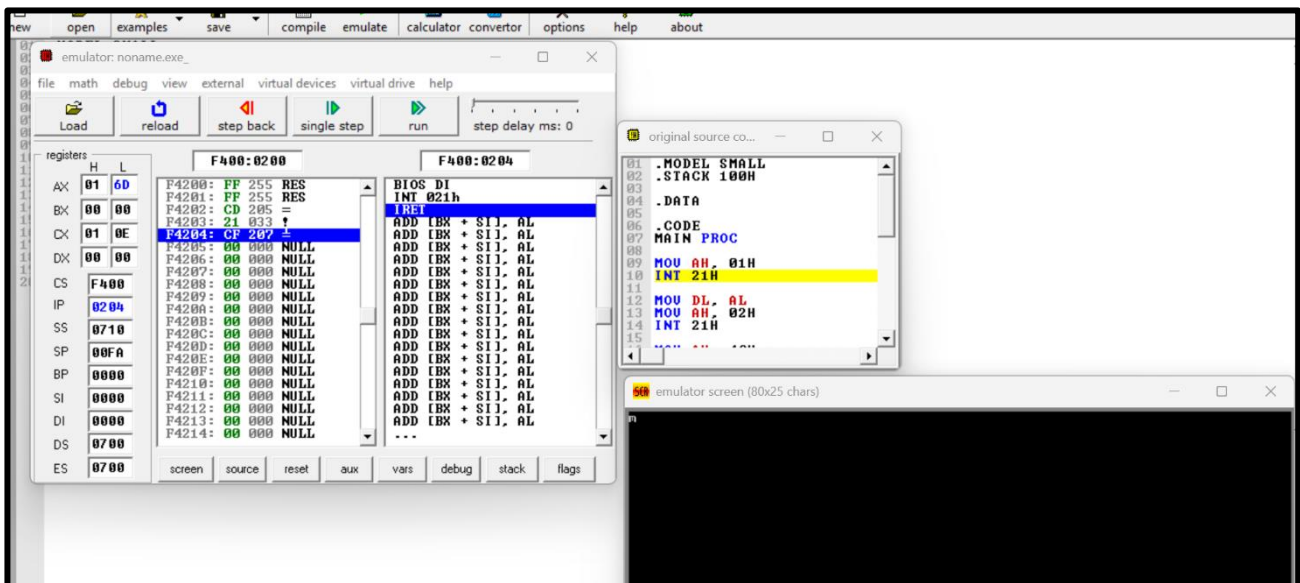
```

file  edit  bookmarks  assembler  emulator  math  ascii codes  help
new  open  examples  save  compile  emulate  calculator  convertor

01 .MODEL SMALL
02 .STACK 100H
03
04 .DATA
05
06 .CODE
07 MAIN PROC
08
09 MOV AH, 01H
10 INT 21H
11
12 MOV DL, AL
13 MOV AH, 02H
14 INT 21H
15
16 MOV AH, 4CH
17 INT 21H
18
19 MAIN ENDP
20 END MAIN

```

Output

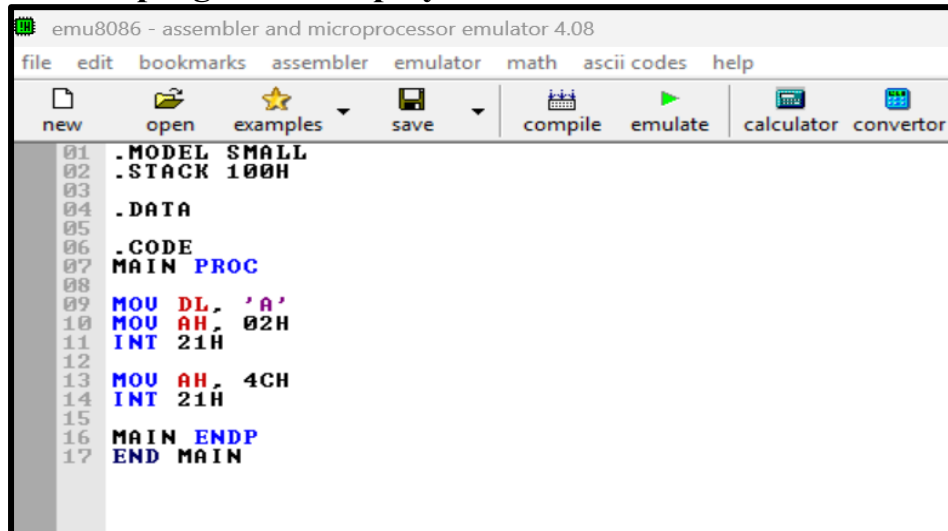


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Task 2

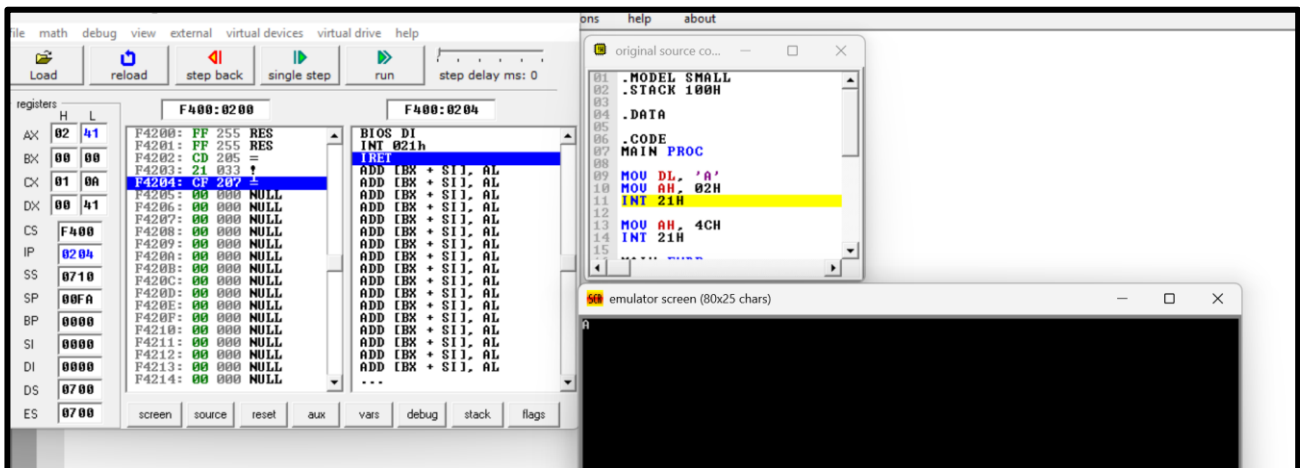
Write a program to display one character on the screen.



```
emu8086 - assembler and microprocessor emulator 4.08
file  edit  bookmarks  assembler  emulator  math  ascii codes  help
new  open  examples  save  compile  emulate  calculator  convertor

01  .MODEL SMALL
02  .STACK 100H
03
04  .DATA
05
06  .CODE
07  MAIN PROC
08
09  MOV DL, 'A'
10  MOV AH, 02H
11  INT 21H
12
13  MOV AH, 4CH
14  INT 21H
15
16  MAIN ENDP
17  END MAIN
```

Output



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Task 3

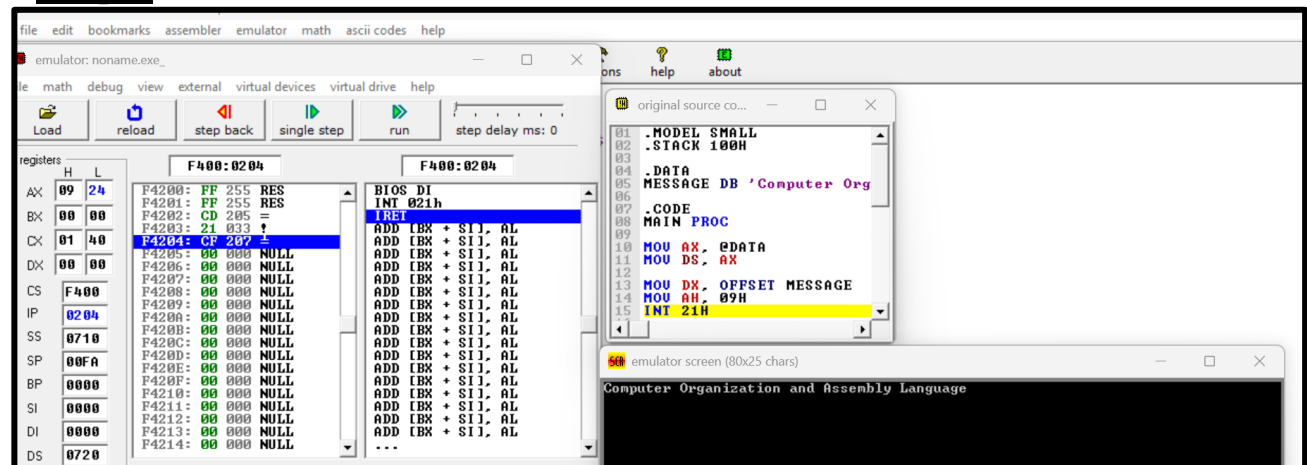
Write a program to display:

Computer Organization and Assembly Language

```
file edit bookmarks assembler emulator math ascii codes help
new open examples save compile emulate calculator convertor options help a

01 .MODEL SMALL
02 .STACK 100H
03
04 .DATA
05 MESSAGE DB 'Computer Organization and Assembly Language$'
06
07 .CODE
08 MAIN PROC
09
10 MOV AX, @DATA
11 MOV DS, AX
12
13 MOV DX, OFFSET MESSAGE
14 MOV AH, 09H
15 INT 21H
16
17 MOV AH, 4CH
18 INT 21H
19
20 MAIN ENDP
21 END MAIN
```

Output



Task 4

Write a program to take four characters from the user and display the entered characters.

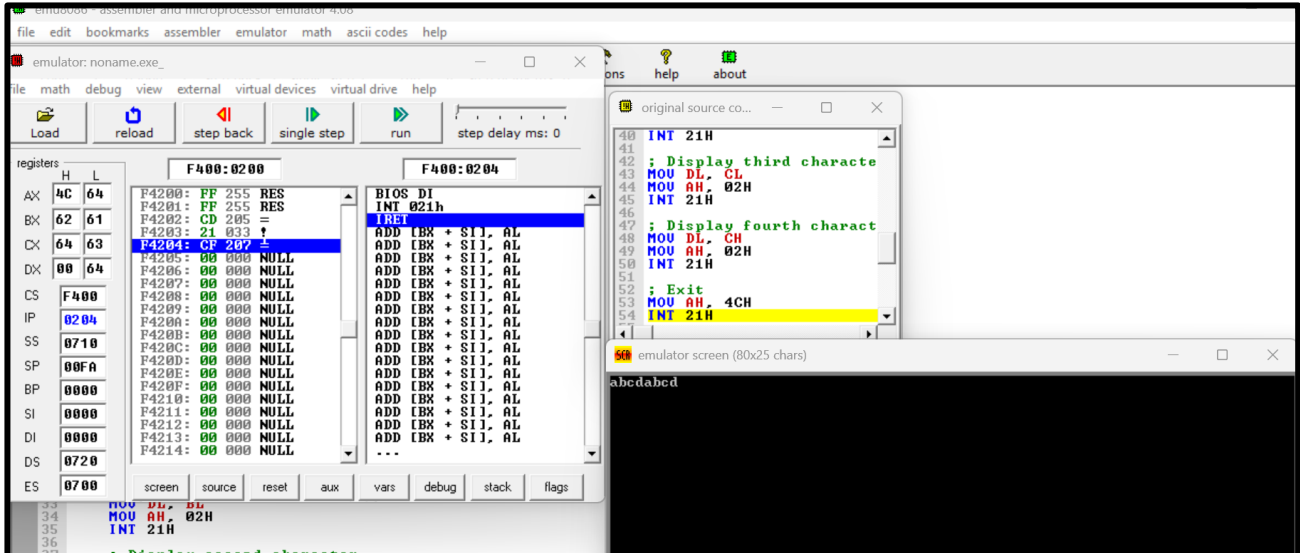
```
new open examples save compile emulate calculator convertor options help

01 .MODEL SMALL
02 .STACK 100H
03
04 .DATA
05
06 .CODE
07 MAIN PROC
08
09     MOV AX, @DATA
10     MOV DS, AX
11
12     ; Take first character
13     MOV AH, 01H
14     INT 21H
15     MOV BL, AL
16
17     ; Take second character
18     MOV AH, 01H
19     INT 21H
20     MOV BH, AL
21
22     ; Take third character
23     MOV AH, 01H
24     INT 21H
25     MOV CL, AL
26
27     ; Take fourth character
28     MOV AH, 01H
29     INT 21H
30     MOV CH, AL
31
32     ; Display first character
33     MOV DL, BL
34     MOV AH, 02H
35     INT 21H
36
37     ; Display second character
38     MOV DL, BH
39     MOV AH, 02H
40     INT 21H
41
42     ; Display third character
43     MOV DL, CL
44     MOV AH, 02H
45     INT 21H
46
47     ; Display fourth character
48     MOV DL, CH
49     MOV AH, 02H
50     INT 21H
51
52     MOV AH, 4CH
53     INT 21H
54
55 MAIN ENDP
56 END MAIN
```

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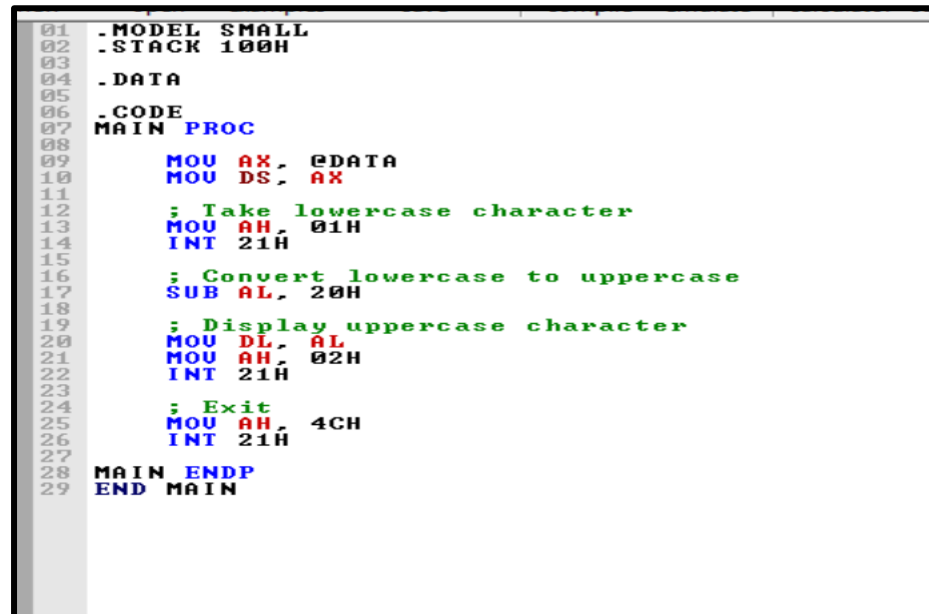
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Output



Task 5

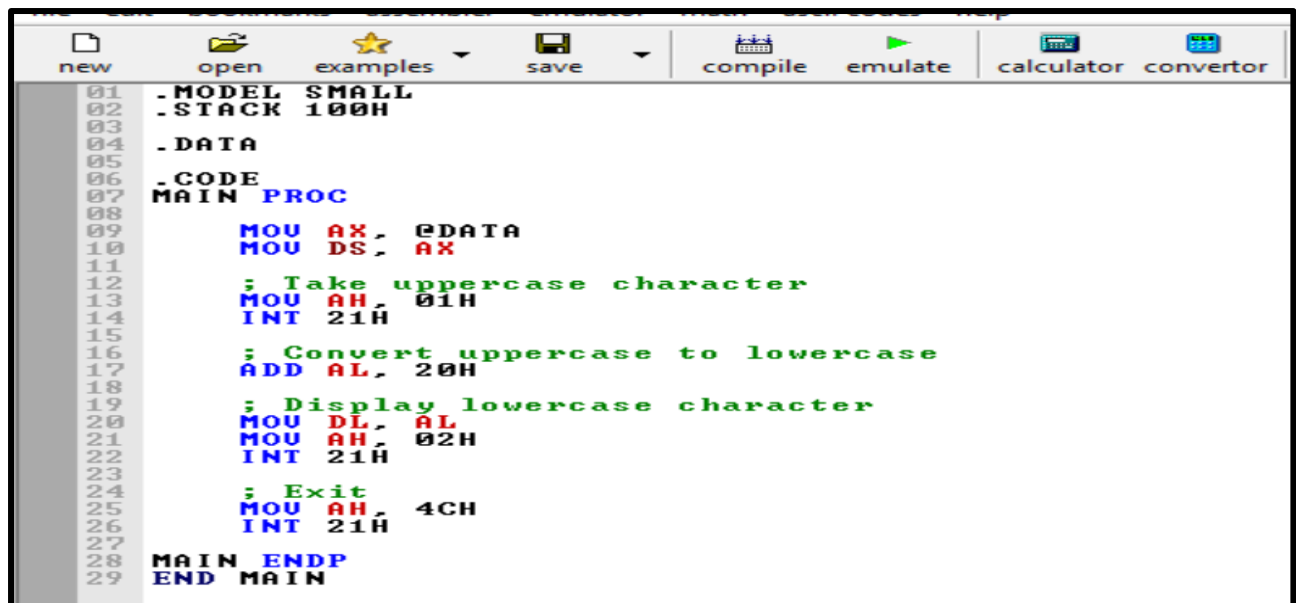
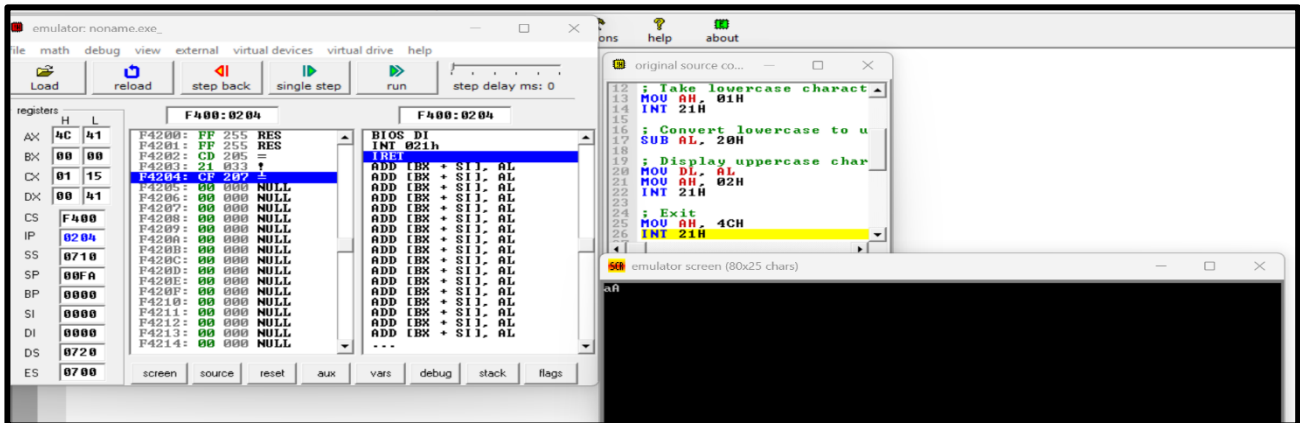
Write a program that takes a lowercase character and converts it into uppercase. Then modify the program to convert an uppercase character into lowercase.



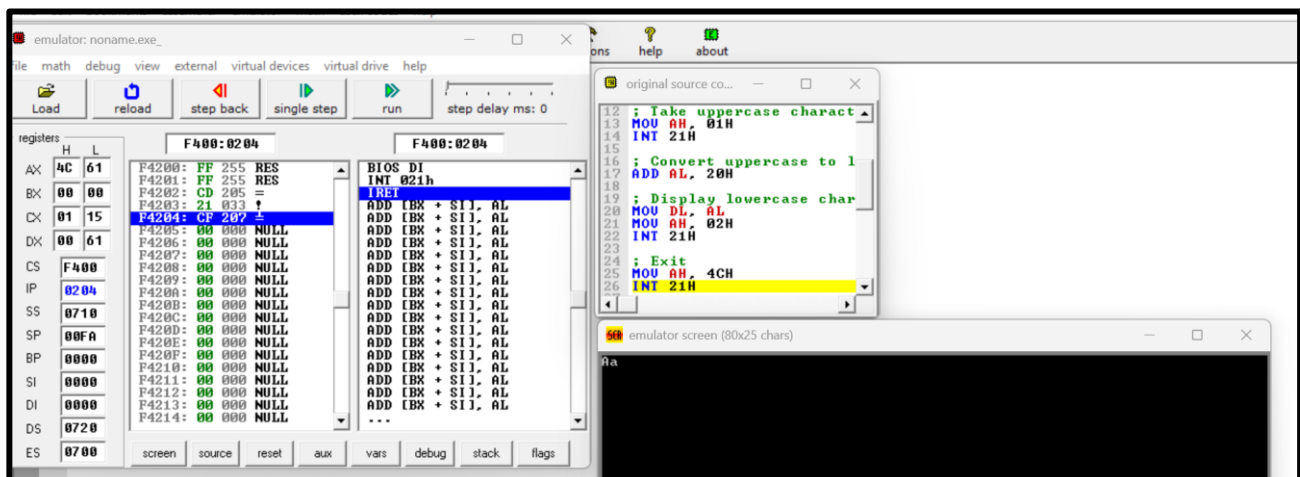
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Output



Output



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Task 6

Write a program that displays:

as a four-character password.

```
01 -MODEL SMALL
02 -STACK 100H
03
04 -DATA
05
06 -CODE
07 MAIN PROC
08
09     MOV AX, @DATA
10     MOV DS, AX
11
12     ; Display first *
13     MOV DL, '*'
14     MOV AH, 02H
15     INT 21H
16
17     ; Display second *
18     MOV DL, '*'
19     MOV AH, 02H
20     INT 21H
21
22     ; Display third *
23     MOV DL, '*'
24     MOV AH, 02H
25     INT 21H
26
27     ; Display fourth *
28     MOV DL, '*'
29     MOV AH, 02H
30     INT 21H
31
32     ; Exit program
33     MOV AH, 4CH
34     INT 21H
35
36 MAIN ENDP
37 END MAIN
```

Output

